

vector sum that includes only the velocity created by the rotation of the blade and that resulting from the forward motion. This definition is shown graphically in Fig. 2.5.1. From the geometry shown in this figure, we can write

$$V_e^2(r, \theta) = (\omega r - V_\infty \sin \alpha_p \sin \theta)^2 + (V_\infty \cos \alpha_p)^2 \quad (2.5.1)$$

Because the propeller angle of attack, α_p , is typically small, we apply the small-angle approximation to this angle only and Eq. (2.5.1) is closely approximated as

$$V_e^2(r, \theta) \cong \omega^2 r^2 + V_\infty^2 - 2\omega r V_\infty \alpha_p \sin \theta \quad (2.5.2)$$

If we now apply the Betz condition according to Goldstein's vortex theory, the total airspeed relative to the blade cross-section is

$$V_b(r, \theta) = V_e \cos \varepsilon_i = \sqrt{\omega^2 r^2 + V_\infty^2 - 2\omega r V_\infty \alpha_p \sin \theta} \cos \varepsilon_i \quad (2.5.3)$$

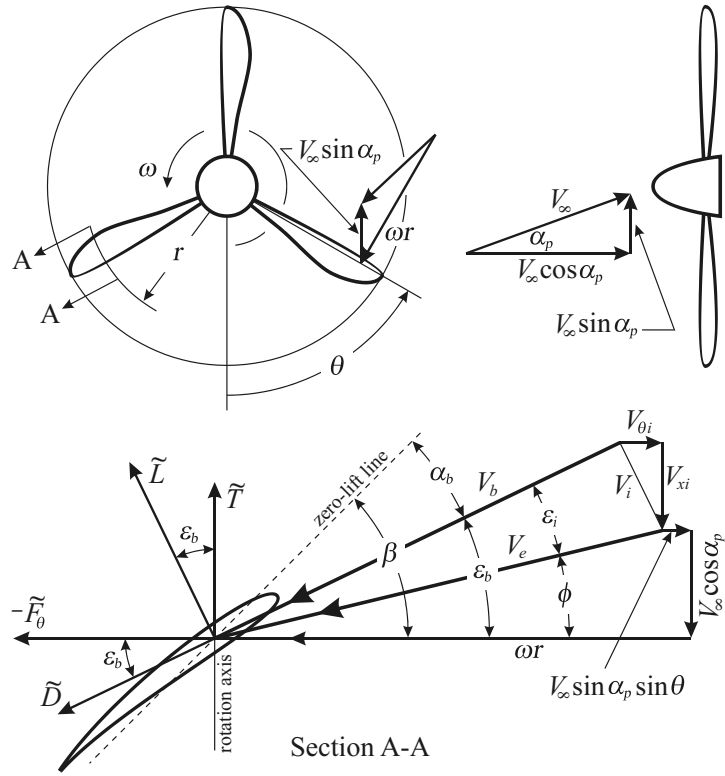


Figure 2.5.1. Effect of propeller angle of attack on the section forces and velocities acting on a rotating propeller blade.